

Twinspark Tyres And Bike Garage — Garage Management System

Full Functional & Data Specification (Single Source of Truth)

Purpose of this document: This is the complete, self-contained specification for the Twinspark Garage Management System. It is meant to be used as the single reference context when generating code (e.g. in Claude Code) — every module, field, business rule, role permission, invoice format, and report is documented here so no additional context should be required.

1. Project Overview

Client: Twinspark Tyres And Bike Garage **Location:** Coimbatore, Tamil Nadu **Business Type:** Premium motorcycle tyre retailer + professional motorcycle service center, serving performance/sports bikes (Yamaha, KTM, BMW, Kawasaki, and other premium brands).

Core Business Lines:

- 1. Selling **Track Tyres**
- 2. Selling **Brand New Tyres** and other motorcycle products
- 3. **Motorcycle Service & Garage Operations**

Other Inventory Sold: Engine Oil, Chains, Chain Sprocket Kits, Brake Parts, Lubricants, Accessories, Other Motorcycle Spare Parts.

Goal: Replace manual inventory/sales/service record-keeping with a single centralized web application covering Inventory, Purchases, Sales, Service Jobs, Billing, Customers, Online Orders, and Reports — improving inventory accuracy and business visibility while reducing manual work.

2. System Modules (Top Level)

#	Module
1	Dashboard
2	Inventory Management
3	Purchase Management
4	Sales Management
5	Service Management

#	Module
6	Job Card
7	Customer Management
8	Vehicle Management
9	Billing & Invoice Management
10	Online Track Tyre Orders
11	Reports
12	User & Role Management

3. Core Data Entities

These are the core objects the system is built around. Field names are suggestions; adapt casing/naming to your stack, but keep the meaning and relationships intact.

3.1 User

Field	Type	Notes
id	UUID/int	primary key
name	string	
mobile_number	string	used for login (or email, TBD)
password_hash	string	
role	enum	ADMIN, SALES_PERSON
is_active	boolean	for disabling accounts
created_at / updated_at	datetime	

3.2 Customer

Field	Type	Notes
id	UUID/int	
name	string	

Field	Type	Notes
mobile_number	string	primary lookup key for repeat customers
address	text	
created_at	datetime	

Customers are created/reused automatically whenever a Sale or Service is recorded against a mobile number. Search-by-mobile-number should auto-fill name/address for repeat customers.

3.3 Vehicle

Field	Type	Notes
id	UUID/int	
customer_id	FK → Customer	a customer can have multiple vehicles
vehicle_number	string	registration number
vehicle_model	string	e.g. "KTM Duke 390"
odometer_reading	integer	captured per service visit (store latest + history)

3.4 Category (for Brand New Tyres only)

Field	Type	Notes
id	UUID/int	
name	string	e.g. "Sports Touring", "Off-Road"

3.5 Brand (for Brand New Tyres only)

Field	Type	Notes
id	UUID/int	
name	string	e.g. "Michelin", "Pirelli", "MRF"

3.6 Inventory Item

Covers: Track Tyres, Brand New Tyres, Engine Oil, Chains, Chain Sprocket Kits, Brake Parts, Lubricants, Accessories, Other Spare Parts.

Field	Type	Notes
id	UUID/int	
item_type	enum	TRACK_TYRE BRAND_NEW_TYRE ENGINE_OIL CHAIN SPROCKET_KIT BRAKE_PART LUBRICANT ACCESSORY OTHER_SPARE_PART
product_name	string	required
category_id	FK → Category	required for Brand New Tyres only; must be null/hidden for Track Tyres
brand_id	FK → Brand	required for Brand New Tyres only; must be null/hidden for Track Tyres
purchase_price	decimal	always stored separately from selling price
selling_price	decimal	always stored separately from purchase price
available_quantity	integer	current stock; auto-maintained, not manually editable outside Purchase/Sales/Service flows
low_stock_threshold	integer	used to trigger dashboard low-stock alerts
created_at / updated_at	datetime	

Business rule reminders:

- Track Tyres → **no** category, **no** brand.
- Brand New Tyres → category **and** brand are mandatory.
- Purchase Price and Selling Price must always be two distinct stored fields — never derived from one another.

3.7 Purchase Entry

Field	Type	Notes
id	UUID/int	
inventory_item_id	FK → Inventory Item	
quantity	integer	

Field	Type	Notes
purchase_price	decimal	price paid per unit for this specific purchase (may differ over time even for the same item)
total_amount	decimal	quantity × purchase_price
purchase_date	datetime	
created_by	FK → User	must be an Administrator

Behavior: On save, `available_quantity` on the linked Inventory Item increases by `quantity`. A full purchase history log must be retained (not overwritten).

3.8 Sale

Field	Type	Notes
id	UUID/int	
customer_id	FK → Customer	
sale_date	datetime	
fitting_required	boolean	tyre fitting flag
fitting_wheel_count	integer	number of wheels fitted (used to calc fitting charge)
fitting_charge_total	decimal	<code>(fitting_wheel_count × ₹300)</code> — always itemized separately from product cost
gst_applicable	boolean	optional
gst_amount	decimal	computed if applicable
discount_applicable	boolean	optional
discount_amount	decimal	flat or % — define one convention
subtotal	decimal	sum of all SaleItem line totals
grand_total	decimal	subtotal + fitting_charge_total + gst_amount – discount_amount
invoice_id	FK → Invoice	
created_by	FK → User	Admin or Sales Person

3.9 Sale Item (line item within a Sale)

Field	Type	Notes
id	UUID/int	
sale_id	FK → Sale	
inventory_item_id	FK → Inventory Item	
quantity	integer	
unit_selling_price	decimal	pulled automatically from Inventory Item's <code>selling_price</code> at time of sale
line_total	decimal	$\text{quantity} \times \text{unit_selling_price}$

Behavior: On sale completion, `available_quantity` on each referenced Inventory Item decreases by the sold `quantity`. Applies uniformly to Track Tyres, Brand New Tyres, and all other inventory categories (oil, chains, sprockets, accessories, etc.) — they all flow through this same Sales module.

3.10 Service Job (Job Card)

Field	Type	Notes
id	UUID/int	
customer_id	FK → Customer	
vehicle_id	FK → Vehicle	
odometer_reading	integer	captured at time of service
service_category	enum	<code>GENERAL_SERVICE</code> , <code>SPECIFIC_SERVICE</code>
selected_general_package_id	FK → General Service Package	if <code>GENERAL_SERVICE</code>
selected_specific_services	list of FK → Specific Service	if <code>SPECIFIC_SERVICE</code> ; multi-select allowed
labour_charge	decimal	entered separately by staff
service_charge	decimal	from General Service package, if applicable

Field	Type	Notes
job_card_rows	list of Job Card Line	see 3.11
service_date	datetime	
invoice_id	FK → Invoice	
created_by	FK → User	

3.11 Job Card Line (the "No | Description | Amount" table)

Field	Type	Notes
id	UUID/int	
service_job_id	FK → Service Job	
line_no	integer	auto-incremented row number
description	string	free text or linked to a service/inventory item
amount	decimal	

Behavior: Staff can add/remove rows freely; system auto-calculates the running total and grand total.

3.12 General Service Package

Field	Type	Notes
id	UUID/int	
name	string	e.g. "Standard General Service"
included_items	list of string	e.g. Oil Change, Water Wash, Standard Inspection, etc. (configurable)
service_charge	decimal	package price

3.13 Specific Service (individually selectable)

Field	Type	Notes
id	UUID/int	
name	string	e.g. Oil Change, Brake Service, Water Wash, Chain Cleaning

Field	Type	Notes
default_charge	decimal	optional default price, editable per job

3.14 Service Inventory Usage

Field	Type	Notes
id	UUID/int	
service_job_id	FK → Service Job	
inventory_item_id	FK → Inventory Item	e.g. Engine Oil, Chains, Brake Pads, Tyres, Accessories
quantity_used	integer	
line_total	decimal	quantity_used × unit selling/cost price (define convention)

Behavior: Reduces `available_quantity` on the Inventory Item automatically. Appears as its own separate section on the Service Invoice (never merged with labour or service charge lines).

3.15 Invoice

Field	Type	Notes
id	UUID/int	
invoice_number	string	unique, sequential
invoice_type	enum	<code>SALES</code> , <code>SERVICE</code>
linked_sale_id / linked_service_job_id	FK	one or the other, depending on type
subtotal	decimal	
additional_charges	decimal	fitting (sales) or labour (service)
gst_amount	decimal	optional
discount_amount	decimal	optional
grand_total	decimal	
invoice_date	datetime	

Field	Type	Notes
pdf_url / print_ready_flag	string/bool	

3.16 Online Track Tyre Order

Field	Type	Notes
id	UUID/int	
customer_name	string	
mobile_number	string	
address	text	
pin_code	string	
quantity	integer	track tyres ordered
payment_screenshot_url	file/image	uploaded by customer
status	enum	SUBMITTED , PAYMENT_VERIFIED , APPROVED , DISPATCHED , REJECTED
submitted_at	datetime	
verified_by / approved_by / dispatched_by	FK → User	
dispatched_at	datetime	

Critical rule: Inventory (Track Tyre stock) is reduced **only** when status changes to **DISPATCHED** — not at submission, payment verification, or approval.

3.17 Courier Label Export

Generated on demand from one or more selected Online Orders (typically **APPROVED** or **DISPATCHED**). Produces a single printable PDF with one label per order, each containing:

- Customer Name
- Mobile Number
- Address
- PIN Code
- Quantity

4.1 Authentication & User Roles

- Two roles only: **Administrator** and **Sales Person**.
- Login via mobile number/username + password (define exact mechanism during build).
- Role determines visible modules and available actions (see permission matrix in Section 6).

4.2 Dashboard

Metrics displayed:

- Current Track Tyre Stock (quantity)
- Total Sales (count and/or amount — clarify with client which is meant)
- Purchase Amount (aggregate, likely current month/period)
- Sales Amount (aggregate)
- Profit → $\text{Profit} = \text{Sales Amount} - \text{Purchase Amount}$
- Current Month Revenue
- Low Stock Alerts — auto-populated list of Inventory Items where $\text{available_quantity} \leq \text{low_stock_threshold}$

Access: Administrator only (Sales Person has no Dashboard analytics access per role rules).

4.3 Inventory Management

- Lists all inventory items across all types (Track Tyres, Brand New Tyres, Engine Oil, Chains, Sprocket Kits, Brake Parts, Lubricants, Accessories, Other Spare Parts).
- Add/Edit inventory item form:
 - If $\text{item_type} = \text{TRACK_TYRE}$ → hide Category and Brand fields.
 - If $\text{item_type} = \text{BRAND_NEW_TYRE}$ → require Category and Brand selection.
 - Always capture Purchase Price and Selling Price as two separate fields.
 - Available Quantity is **read-only** on this screen — it only changes via Purchase Entries, Sales, or Service usage, never edited directly (except perhaps an Admin-only stock correction/audit feature, if desired — flag as an open question).
- Access: Administrator only.

4.4 Purchase Management

- "Add Purchase Entry" form: select Inventory Item → enter Purchase Quantity → enter Purchase Price → Save.

- On save: `available_quantity` increases automatically; a Purchase history record is stored (never deleted/overwritten — full audit trail).
- Purchase History view: filterable list of all past purchase entries (by item, date range, etc.).
- Access: Administrator only.

4.5 Sales Management

- Single unified Sales screen for **all** inventory types (Track Tyres, Brand New Tyres, Engine Oil, Chains, Sprockets, Accessories, etc.).
- Flow:
 1. Enter/select Customer (Name, Mobile Number, Address) — auto-suggest existing customers by mobile number.
 2. Add one or more Sale Items:
 - **Track Tyre:** select item, enter quantity only (no category/brand prompts).
 - **Brand New Tyre:** select Category → select Brand → select specific item → enter quantity. Selling Price auto-pulled from Inventory.
 - **Other items** (oil, chains, sprockets, accessories, etc.): select item, enter quantity, selling price auto-pulled from Inventory.
 3. Optionally mark **Fitting Required** and specify number of wheels → system computes Fitting Charge at ₹300/wheel, shown as a distinct line, never merged into product subtotal.
 4. Optionally apply GST.
 5. Optionally apply Discount.
 6. System computes Subtotal → + Fitting Charges → + GST – Discount → Grand Total.
 7. On completing the sale: inventory auto-deducts for every sold item; a Sales Invoice is generated.
- Access: Administrator and Sales Person (both can create sales).

4.6 Service Management

- Flow:
 1. Capture/select Customer (Name, Mobile Number, Address).
 2. Capture/select Vehicle (Vehicle Number, Vehicle Model) and current Odometer Reading.
 3. Choose Service Type:
 - **General Service:** select a predefined package (e.g. contains Oil Change, Water Wash, Standard Inspection, etc.); package includes a Service Charge; Labour Charge entered separately.
 - **Specific Service:** select one or more individual services (Oil Change, Brake Service, Water Wash, Chain Cleaning, etc.); Labour Charge entered

separately per job.

4. Record Service Inventory Usage: staff select any inventory items consumed (Engine Oil, Chains, Brake Pads, Tyres, Accessories) with quantities — auto-deducted from stock and itemized on the invoice separately from service/labour charges.
 5. Job Card is generated automatically (see 4.7) capturing all of the above plus a free-form line-item table.
 6. Optionally apply GST and/or Discount.
 7. Generate Service Invoice.
- Access: Administrator (full); Sales Person has **no** access to Service Management per role rules (not listed among Sales Person permissions).

4.7 Job Card

- Every Service entry automatically produces a Digital Job Card containing:
 - Customer Details
 - Vehicle Details
 - A line-item table:

No	Description	Amount
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 — staff can add multiple rows freely.
 - Auto-calculated totals.
- Printable/exportable, tied 1:1 with its Service Job and its resulting Invoice.

4.8 Customer Management

- Central customer directory: Name, Mobile Number, Address.
- Auto-created/reused whenever a Sale or Service references a mobile number.
- Should support search/filter by name or mobile number, and show a customer's full sales + service history.

4.9 Vehicle Management

- Tracks vehicles per customer: Vehicle Number, Vehicle Model, Odometer Reading history.
- Supports one customer having multiple vehicles.
- Useful for service history lookups (e.g. "show all past services for KA-01-XXXX").

4.10 Billing & Invoice Management

Both Sales and Service share **one billing engine**, but render two distinct invoice layouts:

Sales Invoice contains:

- Products (line items with quantity, unit price, line total)
- Tyre Fitting Charges (separate line, only if applicable)

- GST (optional)
- Discount (optional)
- Grand Total

Service Invoice contains:

- Service Charges (from General Service package, if applicable)
- Labour Charges (always separate line)
- Inventory Used (separate itemized section — parts/consumables consumed during service)
- GST (optional)
- Discount (optional)
- Grand Total

Strict separation rules:

- Fitting Charges appear **only** on Sales Invoices, never Service.
- Labour Charges appear **only** on Service Invoices, never Sales.
- GST and Discount are optional toggles on **every** invoice type.

4.11 Online Track Tyre Orders

- **Customer-facing (public link, no login required):**
 - Submit: Name, Mobile Number, Address, PIN Code, Quantity of Track Tyres, Payment Screenshot upload.
 - Order enters system with status **SUBMITTED**.
- **Admin/Sales Person workflow:**
 1. View incoming orders (list/queue).
 2. Verify Payment Screenshot → mark **PAYMENT_VERIFIED**.
 3. Approve Order → **APPROVED**.
 4. Dispatch Order → **DISPATCHED** — **this is the only point at which Track Tyre stock is decremented.**
- **Courier Label Export:** select one or more orders (typically approved/dispatched) → export a single printable PDF containing one label per order (Customer Name, Mobile Number, Address, PIN Code, Quantity), ready to attach to courier packages.
- **Access:** Administrator (full workflow) and Sales Person (verify, approve/dispatch, generate labels — per role rules, Sales Person "Can: Verify Online Orders, Dispatch Orders, Generate Courier Labels").

4.12 Reports

Dedicated reports covering:

Report	Suggested Filters/Fields
Inventory Report	current stock by item/category/brand, low-stock flags
Purchase Report	date range, item, quantity, purchase price, supplier (if tracked)
Sales Report	date range, item, customer, quantity, revenue, fitting charges
Service Report	date range, service type, labour charges, inventory used
Revenue Report	period-over-period revenue trends
Profit Report	Sales Amount – Purchase Amount, by period
Stock Movement Report	in/out ledger per inventory item (purchase in, sale out, service usage out, online-dispatch out)

- Access: Administrator only (Sales Person explicitly excluded from Reports).

4.13 User & Role Management

- Administrator can create/deactivate user accounts and assign roles ((ADMIN) / (SALES_PERSON)).
- Access: Administrator only.

5. Consolidated Business Rules (Checklist)

1. Purchase Price and Selling Price are always stored as two separate fields — never one derived from the other.
2. Track Tyres do **not** require Category or Brand.
3. Brand New Tyres **require** both Category and Brand.
4. Stock automatically **increases** after a Purchase Entry.
5. Stock automatically **decreases** after a completed Sale.
6. Stock automatically **decreases** when inventory is consumed during a Service.
7. Every Service generates a Job Card.
8. Every Sale generates an Invoice.
9. Every Service generates an Invoice.
10. GST is optional on every invoice.
11. Discount is optional on every invoice.
12. Tyre Fitting Charges appear **only** on Sales invoices (never Service).
13. Labour Charges appear **only** on Service invoices (never Sales).
14. Fitting Charge rate: **₹300 per wheel**, always itemized separately from product cost.

- 15. Online Orders reduce inventory **only** after Dispatch — not at submission, payment verification, or approval.
- 16. Profit is calculated as: **Profit = Sales Amount – Purchase Amount.**
- 17. Low-stock items automatically surface on the Dashboard (threshold-based).

6. User Roles & Permission Matrix

Module / Action	Administrator	Sales Person
Dashboard (analytics)	✓	✗
Inventory Management	✓	✗
Purchase Management	✓	✗
Sales — Create Sale / Generate Sales Invoice	✓	✓
Service Management	✓	✗
Reports (all types)	✓	✗
Online Orders — View	✓	✓
Online Orders — Verify Payment	✓	✓
Online Orders — Approve	✓	✓ (per "Verify Online Orders / Dispatch Orders")
Online Orders — Dispatch	✓	✓
Online Orders — Courier Label Export	✓	✓
Settings	✓	✗
User & Role Management	✓	✗
Customer Management	✓	✓ (implicitly, via Sales flow)
Vehicle Management	✓	✗ (tied to Service, which Sales Person cannot access)

⚠ **Open question to confirm with client:** The PRD states Sales Person can "Verify Online Orders" and "Dispatch Orders" but does not explicitly mention "Approve."

Recommend clarifying whether Approve is Admin-only or shared with Sales Person before building the permission gate.

7. Invoice Numbering & Format Notes

- Recommend a unique, sequential invoice number scheme, e.g. **TW-S-000123** (Sales) and **TW-J-000123** (Service/Job) to distinguish invoice types at a glance — confirm preferred prefix with client.
 - Invoices should be printable and exportable as PDF.
 - Every invoice must clearly separate: Subtotal → Additional Charges (Fitting or Labour, as applicable) → GST (if applicable) → Discount (if applicable) → Grand Total.
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8. Project Scope

In Scope

Dashboard, Inventory Management, Purchase Management, Sales Management, Service Management, Customer Management, Vehicle Management, Billing & Invoice Generation, Job Card Management, Online Track Tyre Orders, Courier Label PDF Export, Reports, User & Role Management.

Out of Scope (unless separately requested)

- Third-party accounting software integration
 - SMS/WhatsApp notifications to customers
 - Multi-branch / multi-location support
 - Payment gateway integration (online orders currently rely on manual screenshot verification)
 - Mobile native apps (this is a web application)
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9. Open Questions / Assumptions to Confirm Before Build

1. Login mechanism: mobile number + password, or email-based? Any OTP requirement?
2. Discount: flat amount or percentage (or configurable per invoice)?
3. Should Administrators be able to manually correct/audit stock quantities outside the Purchase/Sale/Service flows (e.g. for damage, loss, stock-take)?
4. Supplier tracking on Purchases — needed now or future enhancement?
5. Exact wording/threshold logic for "low stock" per item (fixed number vs. percentage of typical stock).

6. Should Sales Person's "Approve" permission on Online Orders be enabled or restricted to Admin only (see Section 6 note)?
 7. GST rate(s) — single flat rate or configurable/multiple slabs?
 8. Invoice numbering format/prefix preference.
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End of specification. This document consolidates and supersedes the original PRD drafts for the purpose of development context.